

# AssetFlow

Smart Asset Management & Resource Allocation Platform

Hackathon Submission

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## Abstract:

AssetFlow is a centralized platform designed to streamline asset allocation, booking, tracking, approval workflows, and inventory management within educational institutions and organizations. The platform eliminates manual tracking processes by providing a unified digital system for requesting, approving, issuing, and returning assets while maintaining complete auditability.



# Problem Statement, Objectives & Project Overview

## Problem Statement

- Spreadsheets/manual tracking
- Asset misplacement
- Double bookings
- Poor inventory visibility
- Slow approval workflows

## Objectives

- Digitize asset management
- Simplify bookings
- Improve visibility
- Enable role-based admin
- Maintain audit trails
- Reduce overhead

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## Project Overview

**Project Name:** AssetFlow

**Objective:** Smart asset management and resource allocation for efficient tracking, booking, and administration.

### Key Features:

- Centralized asset inventory
- Booking and approval workflow
- Role-based access and admin controls
- Audit trails and status visibility

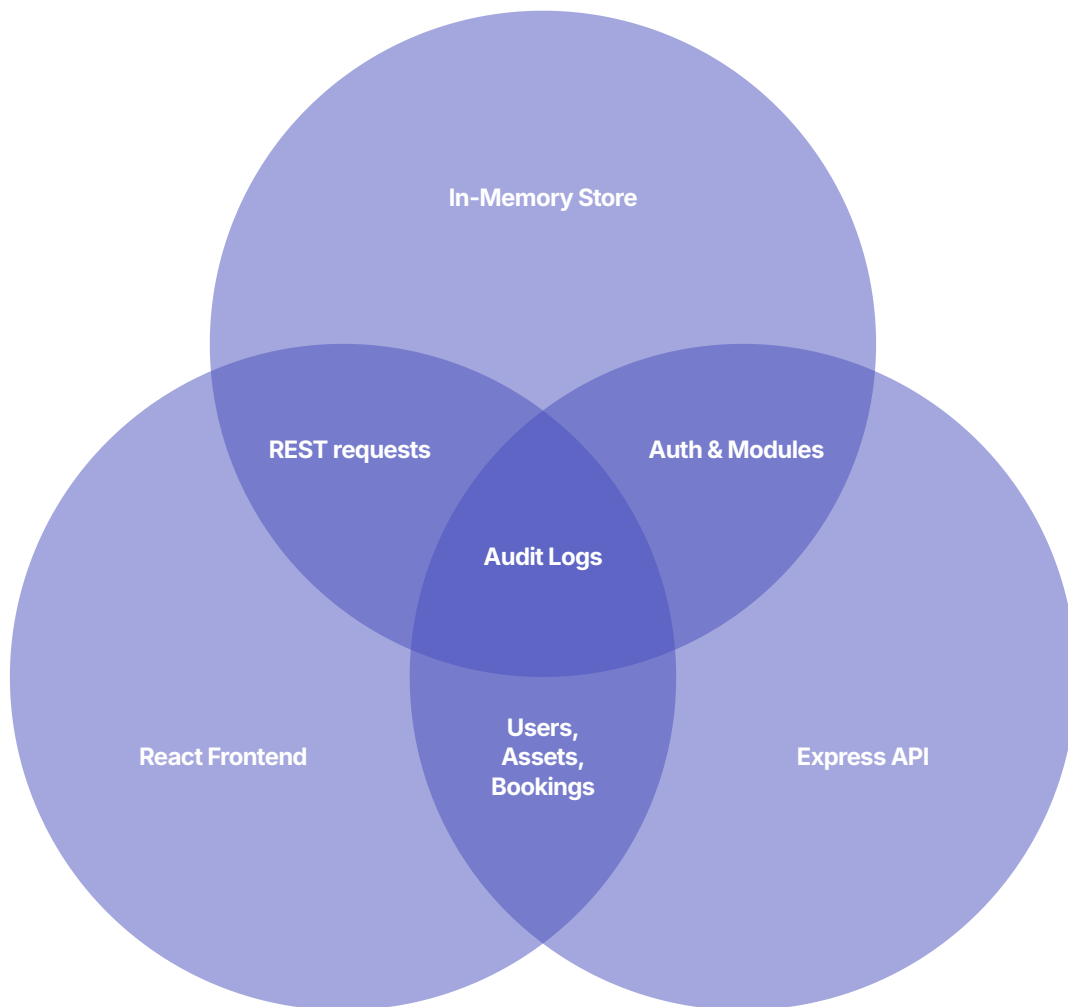
**Technology Stack:** Modern web application stack with database-backed asset tracking, workflow automation, and role-based permissions.

**Team:** Aaditya S Khichar and Naman Awana

# Architecture, Components & Data Model

AssetFlow uses a modular architecture that separates the user interface, API layer, and data storage, while core services handle authentication, booking workflows, and auditing.

## System Architecture



Three layers work together: the React frontend handles the UI, the Express API processes requests, and the in-memory data store manages application data.

# Core Components & API Endpoints

AssetFlow exposes a focused API for authentication, asset management, bookings, and auditing.

## Core Components

### Authentication & Assets

- User login, registration, and access control
- JWT token management
- Asset creation, tagging, and status tracking

### Booking Module

- Booking requests and approvals
- Calendar scheduling
- Return management

### Audit Module

- Audit trail logging
- Asset movement tracking
- Compliance reporting

## API Endpoints

|                |                                                                        |
|----------------|------------------------------------------------------------------------|
| Authentication | POST /auth/register, POST /auth/login                                  |
| Assets         | GET /assets, POST /assets, PUT /assets/:id, DELETE /assets/:id         |
| Bookings       | GET /bookings, POST /bookings, PUT /bookings/:id, DELETE /bookings/:id |
| Audit          | GET /audit-logs                                                        |

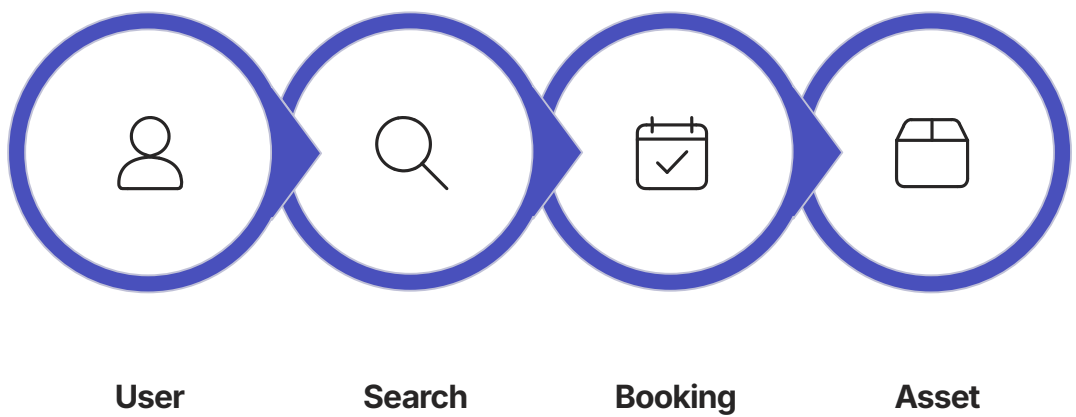
# Data Model & Database Schema

The database schema supports AssetFlow workflows with four core tables and clear relationships between users, assets, bookings, and audit logs.

## Data Model

| Table     | Fields                                                         |
|-----------|----------------------------------------------------------------|
| Users     | id, name, email, role, passwordHash                            |
| Assets    | id, name, category, quantity, status, tag, location            |
| Bookings  | id, userId, assetId, quantity, startDate, endDate, status      |
| AuditLogs | id, userId, action, entity_type, entity_id, timestamp, details |

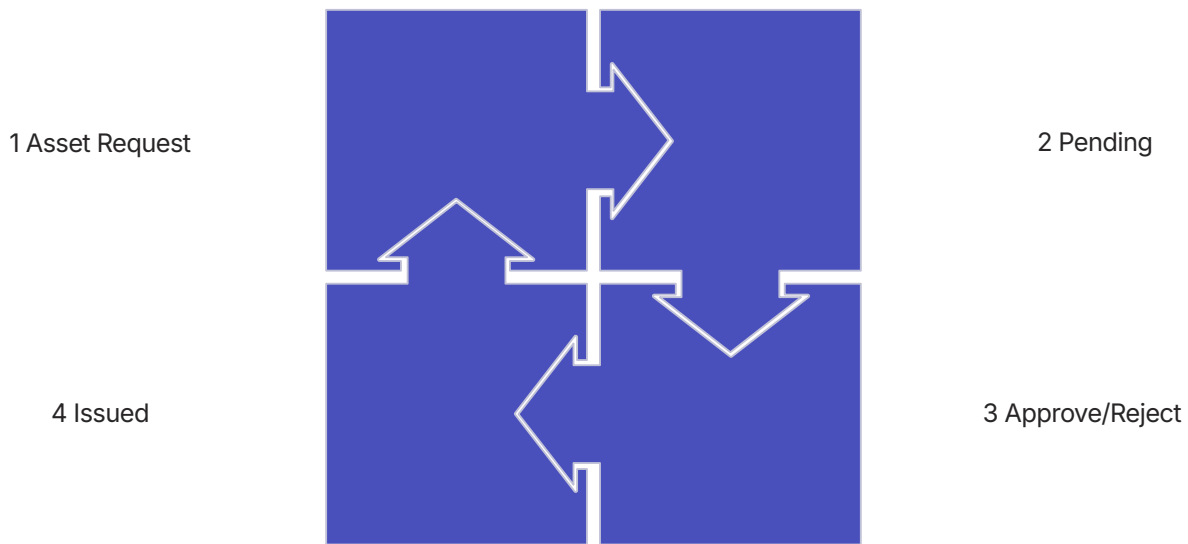
## Entity Relationships



This diagram illustrates the core entity relationships, where a user initiates a booking for a specific asset, forming the basis of AssetFlow's operational logic.

# Booking Lifecycle & Key Features

## Booking Lifecycle

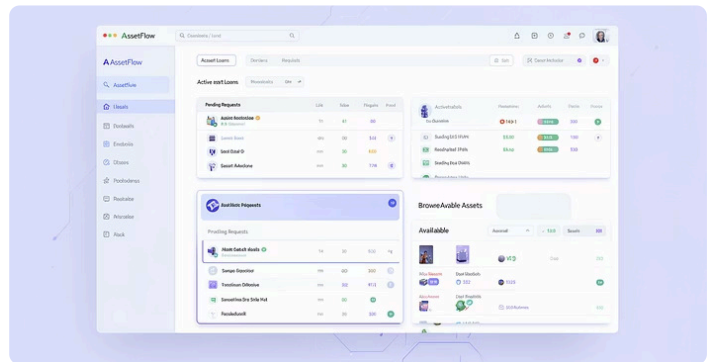
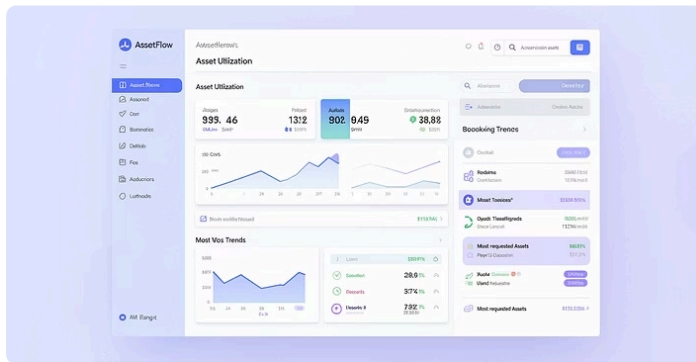
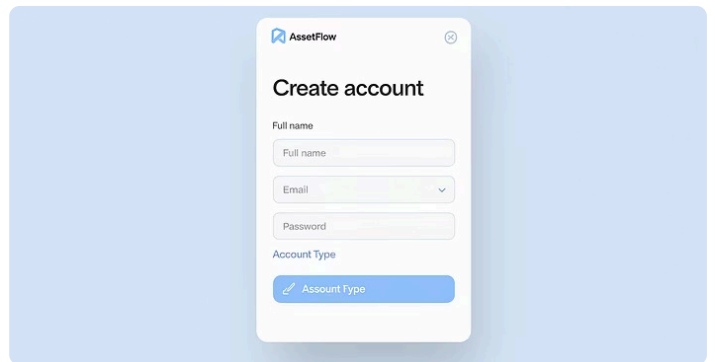
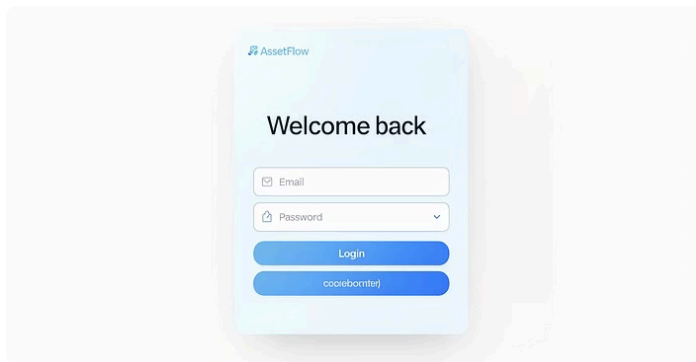


## Key Features

- Secure Authentication: JWT-based user authentication
- Asset Discovery: Browse and search available assets
- Smart Booking System: Calendar-based booking interface
- Approval Workflow: Multi-level approval process
- Role-Based Dashboards: Customized views for different user roles
- Audit Logging: Complete history of all transactions

# User Interface & Screenshots

AssetFlow features an intuitive interface for both regular users and administrators.



# Design Decisions, Future Scope & Conclusion

## Major Design Decisions

- **React + Vite:** fast development, modern tooling, reusable components
- **Express.js:** lightweight backend with strong REST API support
- **In-memory storage:** simple, quick to build, ideal for hackathon constraints

## Future Scope

Expanding AssetFlow to support persistence, automation, and smarter asset management.

- **PostgreSQL integration**
- **RFID-based asset tracking**
- **Mobile application**
- **Email notifications**

## Conclusion

AssetFlow delivers a centralized digital solution for asset lifecycle management. It improves efficiency, transparency, and accountability while reducing manual errors and approval bottlenecks.